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MILITARY SPECIFICATION

TRUCKS, 10 TON, 6X6, M121A1 SERIES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers various tactical vehicles diesel driven, of the 10 Ton, 6X6, M121A1 series as used by the U.S. Armed Forces.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-C-520 – Coating, Underbody (For Motor Vehicles).

Military

MIL-P-514 – Plates; Identification, Instruction, and Marking Blank.
MIL-R-3065 – Rubber, Fabricated Parts.
MIL-E-52396 – Engine, Diesel: 8-Cylinder V-Type 300 H.P. (V8-300).
MIL-V-62038 – Vehicles, Wheeled: Preparation for Shipment and Storage of.

STANDARDS

Federal

Federal Standard No. 595 – Colors.

Military

MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 – Identification Marking of U.S. Military Property.
MIL-STD-193 – Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-210 – Climatic Extremes for Military Equipment.

FSC-2320

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MIL-STD-417 – Rubber Compositions Vulcanized General Purpose, Solid Symbols and Tests.
MIL-STD-461 – Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 – Identification Marking of Combat, and Tactical Transport Vehicles.
MS35000 – Battery, Storage, Lead-Acid, Waterproof.
MS35647-7 – Padlock, Key Operated.
MS53050 – Plate, Good Driver, Adhesive Backed, for Wheeled Vehicles.

Drawings

Army

D6566675 – Bracket, Liquid Container.
7071880 – Fittings.
7346922 – Bracket, Pioneer Tools.

PUBLICATIONS

Engineering Parts Lists

8736670 – Chassis, Truck: 10 Ton, 6X6, Diesel Engine, W/Winch, M121A1.
8736671 – Truck, Cargo: 10 Ton 6X6, Diesel Engine, W/Winch, M125A1.
8736634 – Truck Tractor: 10 Ton 6X6, Diesel Engine, W/Winch, M123A1C.

Master Lists

8736670 – Master List of Supplementary Quality Assurance Provisions for Chassis, Truck: 10 Ton, 6X6, Diesel Engine, W/Winch M121A1.
8736671 – Master List of Supplementary Quality Assurance Provisions for Truck Cargo: 10 Ton 6X6, Diesel Engine, W/Winch, M125A1.
8736634 – Master List of Supplementary Quality Assurance Provisions for Truck, Tractor: 10 Ton 6X6, Diesel Engine W/Winch, M123A1C.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), a preproduction vehicle(s) of the complete vehicle to be furnished under this specification shall be produced prior to the manufacture of fabrication of vehicles in quantity. The vehicle(s) when complete shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacture from production facilities and tooling under the supply contract.

3.2 First production vehicles. The vehicles selected, shall be fully representative of vehicles proposed to be furnished under the contract, shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

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3.2.1 *Workmanship and reliability.* The workmanship shall be of quality to assure that the vehicle and components thereof are free from any defects that might compromise, limit, or reduce the vehicle or component capabilities herein expressed. Vehicles made in accordance with this specification shall be subjected to examinations and tests by the Government to prove vehicle capabilities to perform in accordance with the following:

(a) Vehicle shall require only scheduled operators and organizational maintenance during the first 3,000 miles of normal operation.

(b) Vehicle shall require no replacement or major overhaul of any major automotive component during 10,500 miles of normal operation.

3.2.1.1 *Reliability.* All reliability requirements are based on vehicle being capable of traveling the prescribed distance under the following operating conditions.

(a) 30 percent of distance on paved roads either concrete or asphalt of any combination of the two.

(b) 39 percent of distance on level gravel or dirt roads.

(c) 30 percent of distance on hilly cross-country terrain.

(d) 1 percent of distance on Belgian block.

3.3 **Materials.** All materials shall be free from all defects and imperfections that might affect function of finished product.

3.3.1 *Qualified products.* With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed (by part or drawing number as QPL items) on the QPL for the referenced specification or which have been approved for inclusion on such lists. Contractor's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3.2 *Ozone resistance.* Rubber products procured under requirements of MIL-R-3065 are to be Ozone resistant to the degree specified by the C^1 requirements of MIL-STD-417. Rubber components controlled by detail specifications shall be Ozone resistant only when required by the specification referenced on the drawing.

3.4 **Construction.** Vehicle, components, sub-assemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with drawings listed or referred to in Engineering Parts Lists (EPL's) 8736670 M121A1, 8736671 M125A1, and 8736634 M123A1C. All parts, sub-assemblies, and assemblies requiring identification shall be identified in accordance with MIL-STD-130.

3.4.1 *Seals.* When vehicle is fording, the bearing seals shall be capable of preventing the entrance of foreign matter into bearings, which are exposed to contamination during these operations. Water contamination of wheel bearing lubricant not to exceed 2 percent during fording operations. All bearings seals shall prevent the leaking of lubricants from the bearings.

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3.4.2 *Special kits.* When specified (see 6.2), special kits shall be furnished as required for vehicles operating under specific conditions.

3.4.2.1 *Fording kit.* When installed in accordance with applicable drawings the fording kit shall not interfere with operation of vehicle. The vehicle shall ford fresh or salt water up to depths of 78 inches and remain in water for 15 minutes without damage to vehicle.

3.4.2.2 *Arctic kit.* The arctic kit shall be capable of being installed during arctic operations, when installed kit shall not interfere with vehicle operation and shall provide protection for the operating personnel during arctic operations down to minus 65°F.

3.4.2.3 *Personnel heater kit.* The personnel heater shall provide comfort for operating personnel while operating in extreme temperatures down to minus 25°F. The heater shall also be provided with defrosters for keeping windshield free from frost and icing.

3.4.3 *Balancing.* When installed on the vehicle, wheels, hubs, and brake drums shall be in balance, adequate to prevent front wheel vibration and tire wear.

3.4.4 *Brush guards.* Each vehicle shall be equipped with guards, to protect radiator, front lamps and vulnerable engine components from damage when operating in brush areas or similar hazards.

3.4.5 *Controls.* The controls shall all be located for driver convenience and shall include light and dimmer switches, starting switch, hand throttle, and windshield wiper controls. Means shall be furnished to insure positive settings and prevent accidental actuation of controls during fording or other difficult operations. The hand throttle control shall be of distinctive shape so it can be located readily by feel when submerged during fording, and shall permit positive setting for any throttle opening. The vehicle shall include provisions for subsequent installation of controls furnished as part of fording and arctic modification kits.

3.4.6 *Seats.* The driver's seat shall be adjustable and free from interference. Seating shall be provided for one or more men, or when practicable for two or more men.

3.4.7 *Windshield assembly.* The windshield assembly shall be capable of being folded forward to a position level with the engine hood. The adjusting mechanism shall permit windshield to be opened and held in any position up to 15 degrees from horizontal.

3.4.8 *Top installation.* Provisions shall be incorporated to facilitate subsequent installation of hard or soft top without any alteration, and for use when personnel is wearing arctic clothing.

3.4.9 *Ventilation.* Provisions shall be made in the open cab to insure ventilation when operating under extreme high temperatures.

3.4.10 *Tops.*

3.4.10.1 *Soft top.* The soft top, and components shall permit ready and rapid removal, or replacement, and all such parts shall be capable of being stowed within the cab.

3.4.10.2 *Hard top.* The hard top shall be capable of being knocked down for shipment. The hard top shall be capable of being installed readily on open cab by troops in the field, or removal from open cab and substitution of a soft top. Rear windows shall slide up and down or to the side without interference.

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3.4.11 *Fuel and air tanks.* Provisions shall be made to assure the internal cleanliness of the fuel and air tanks prior to initial fueling of vehicle. When installated in accordance with applicable drawings the fuel tank(s) shall to the maximum extent be free from vibration and shall be free from leakage.

3.4.11.1 *Fuel and air lines.* Prior to installation on the vehicle fuel, air lines, and connections shall be internally clean. All fuel, air lines, connections, and tanks shall be free from leakage. All lines shall be protected from vibration or flexing that may cause chaffing and failure of tubing.

3.4.12 *Air cleaner.* Air cleaner when installed in accordance with applicable drawings, shall provide maximum protection against contamination of the air entering the engine intake.

3.4.13 *Exhaust system.* The exhaust system when installed in accordance with applicable drawings, shall be leakproof, and shall not interfere with power plant removal.

3.4.14 *Steering power-assist.* Power-assist steering mechanism, of hydraulic type, shall be furnished. It shall to the maximum practicable extent, provide features of directional stability inherent in the manually-operated steering mechanism. It shall permit manual steering in the event of failure of the hydraulic system, and effective at all vehicle speeds.

3.4.15 *Compressed-air devices.*

3.4.15.1 *Air compressor.* Air compressor intake shall be connected to the engine air cleaner, at a location, permitting use of common air intake when provided for deep water fording operations as well as normal operation.

3.4.15.2 *Air reservoir.* Air reservoirs shall be furnished and installed with safety valve, low pressure warning indicator, pressure gage, connection drain cocks, and piping connection to compressor.

3.4.15.3 *Air-supply valve.* One air-supply valve shall be furnished, accessibly located and equipped to fit tire inflation hose.

3.5.15.4 *Trailer air-connectors.* Air connectors, 2 located at the front and 2 at the rear, and two connections in back of cab on M123A1C, and equipped with dummy caps chained to the vehicle to prevent loss shall be furnished. All connectors shall be equipped with shut-off cocks.

3.4.16 *Slave receptacle kit.* The slave kit furnished and installed at R.H. corner of cab shall be used for starting the engine.

3.5 **Performance.** A complete vehicle, loaded with the rated payload, operating personnel, and serviced shall perform as specified herein. Vehicle, serviced and equipped for existing climatic conditions shall perform as specified herein without special equipment.

3.5.1 *Environmental.* The vehicle shall operate in ambient air temperature of plus 115° to minus 25°F, and with special equipment installed, at temperature ranges of minus 25° down to minus 65°F. The complete vehicle when in storage shall withstand climatic extremes as specified in MIL-STD-210, without deterioration that may cause failure of any component part of the vehicle.

3.5.2 *Power package and power train.* When power package and power train are installed in accordance with applicable drawings, they shall function throughout all gear and speed ranges without loss of lubricants, and damage that may cause failure of vehicle.

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3.5.3 *Engine.* Engine conforming to MIL-E-52396 shall be furnished, and when installed shall provide vehicle performance specified herein.

3.5.3.1 *Engine governor.* The governor furnished shall limit the maximum engine speed to 3000 plus or minus 20 rpm at full load operation and 3300 maximum at no load. After setting, the governor shall be sealed to prevent any subsequent tampering with governor.

3.5.3.2 *Auxiliary speed control assembly for use with power take off operation.* The speed control assembly shall limit the speed of the engine to the desired speed dependent upon the vehicle application.

3.5.3.3 *Cooling system.* The cooling system shall maintain coolant temperatures between 160°F and 210°F, under all service conditions of vehicle operation (see Table I).

3.5.4 *Fuel system.* The vehicle, loaded with cross-country payload, and when vehicles are ascending and descending 60 percent longitudinal grades in forward and reverse gear, the fuel system shall supply fuel to the engine without failure. The system shall supply fuel to engine when vehicle is operating on 20 percent side slopes, with either side of vehicle up slope.

3.5.4.1 *Fuel tanks.* With the vehicle loaded to gross vehicle weight, the fuel tank(s) capacity shall be such that the vehicle will operate for not less than 350 miles at 30 miles per hour.

3.5.5 *Transmission.* The transmission when installed in accordance with applicable drawings, shall provide the maximum ease in gear shifting, and shall transmit the maximum gross torque of the engine throughout all gear reductions, under all conditions of vehicle operation.

3.5.5.1 *Power take-off mechanism.* When specified (see 6.2), the power take-off mechanism, for transmission mountings, shall be of the two forward speed and one reverse speed, double-ended drive type to provide for unobstructed front or rear drive, or combination of both, as required. Unit shall provide minimum practicable difference in component parts with maximum interchangeability of parts, between power take-off mechanisms. This mechanism shall be capable of transmitting the maximum gross torque of the engine as developed through maximum intervening gear reduction. The power take-off shall provide for front drive only, rear drive only, or the combination of front and rear drive.

TABLE I. *Load allowance (pounds)*

Type	Model	Type of Serivce	Gross load incl. body wt. allow.	Class		Towed (pintle) load
				Gross weight w/o winch	Gross weight with winch	
Cargo	M125A1	Cross-country	20,000	52,100	53,600	35,000
		Limited cross-country	20,000	52,100	53,600	25,000
		Highway	30,000	62,100	63,600	50,000

			5th wheel load	Gross weight one winch	Gross weight two winches	Semitrailer incl. 5th wh. load
Truck Tractor		Limited cross-country	30,000	60,650	62,850	110,000
		Highway	35,000	65,650	67,850	120,000
	M123A1C	Cross-country	30,000	59,100	--	110,000

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3.5.5.2 *Two-speed transfer case.* The transfer case shall be of two-speed type with high and low ratio, incorporating power take-off mechanism, and shall be operable with vehicle in motion or at rest.

3.5.6 *Front and rear axles.* The front and rear axles shall be of the multiple reduction gear type, with the rear axles having the final reduction in the axle housing bowl. Unless otherwise specified by the procuring activity, axles shall be of a design that the wheel will not be released in the event of the axle shaft failure. Maximum practicable interchangeability shall be maintained between front and rear axle components. Tread of front and rear axles shall be approximately the same.

3.5.7 *Cramping angle.* Front wheels shall have cramping angle of not less than 35 degrees at the wheel on the inside of the turning circle. Axle stops shall be provided to allow the maximum safe cramping angle.

3.5.8 *Differentials.* When specified (see 6.2), automatic means for permitting differences in speed between the axles of bogie shall insure driving of either axle in the event one has lost its traction.

3.5.9 *Bogie.* A four-wheel rear bogie of the fully articulated, parallelogram, balanced spring suspension type shall be furnished complete with axles, springs, torque-rods and other necessary parts. Stops shall be provided to prevent unnecessary movement. A differential shall be provided between the two rear (bogie) axles. When the vehicle is fully equipped, loaded with rated cross-country payload, and tire chains installed on front and rear wheels; no permanent deformation of the bogie assembly or chassis frame shall occur when blocks each 12 inches high are placed under any wheel or combination of two or more wheels. Compliance with flexibility requirement shall be demonstrated by all bogie wheels maintaining traction whether contacting the ground or any combination contacting the ground with the others contacting the blocks.

3.5.10 *Braking system.* The air braking system shall safely control vehicle fully equipped and loaded to gross vehicle weight, and shall meet braking and holding requirements specified herein.

3.5.10.1 *Service brakes.* The vehicle, fully equipped and loaded to gross vehicle weight shall be decelerated, stopped, held and controlled when descending grades up to 60 percent. On relatively level, dry, hard-surfaced roadway, application of service brakes shall bring the vehicle to complete stop, from a speed of 20 mph within 50 feet (when loaded) and 40 feet (unloaded) from point of brake application.

3.5.10.2 *Hand-control valve.* One hand-control valve shall be installed on steering column, to provide independent application of towed-load air brakes; with stop-light switch or switches arranged so that the vehicle and towed-load stop light will be actuated by either the foot or hand air-brake control.

3.5.10.3 *Parking brake.* The handlever operated, parking brake shall be provided in addition to the service brakes, and mounted on the rear of the transfer case where it shall be effective on the rear (bogie) wheels. The vehicle loaded with highway payload specified in Table I, less towed load, the parking brake when applied shall hold the vehicle motionless when headed up and down a 40 percent grade.

3.5.10.4 *Differential brake system.* The differential brake system control assembly and associated components shall be mounted in the cab, to permit the driver to selectively apply braking action to either the left hand forward-rear and rear-rear axle brakes, or the right hand forward rear and rear-rear axle brakes.

3.5.10.5 *Automatic breakaway safety device.* The automatic breakaway device shall be on all vehicles to provide for automatic brake application of brakes on towed vehicles.

3.5.11 *Operating conditions.*

3.5.11.1 *Cross-country operation.* The vehicle shall transport the rated cross-country payload and towed-load over unimproved roads, trails, open fields, hills, and rough cross-country terrain (see table I).

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3.5.11.2 *Limited cross-country operation.* The vehicle shall transport the rated limited cross-country payload and towed load over reasonably firm terrain other than highways or roads.

3.5.11.3 *Highway operation.* The vehicle shall transport the rated highway payload and towed load over prepared roads. Performance shall be demonstrated on smooth, dry, relatively level, hard-surfaced roadway.

3.5.11.3.1 *High speed.* The vehicle loaded with rated highway payload shall operate at a high speed of not less than 40 miles per hour (mph).

3.5.11.3.2 *Low speed.* The vehicle shall operate at a low speed of not more than 2.50 mph with the engine operating in the speed range between 1800 rpm and 2100 rpm.

3.5.11.4 *Water fording operation.* The vehicle shall transport the rated cross-country payload and towed load when fording hardbottom crossings, in fresh or salt water, of at least 30 inches depth, without requiring the addition of special equipment or adjustments.

3.5.12 *Gradeability.*

3.5.12.1 *Longitudinal grades.* The vehicle shall negotiate not less than 1 3/4 percent grade at a minimum of 27 mph when loaded with cross-country payload and towed load. Vehicle shall negotiate 60 percent grades without towed load, without overheating stalling of engine or slipping of vehicle. Vehicle performance shall be demonstrated on smooth, dry, hard-surfaced roadway. The M123A1C and M123A1 vehicles shall be capable of negotiating the grades, when loaded in accordance with the gross weight specified in table I for the M125A1 cargo vehicles.

3.5.12.2 *Side slopes.* The vehicles fully equipped and loaded with cross country payload shall operate satisfactorily on side slopes right and left up to 20 percent without stalling, overheating or upsetting.

3.5.13 *Lower fifth wheel and equipment.* The lower fifth wheel and equipment when installed in accordance with the applicable drawings shall perform its individual function as used for the intended purpose.

3.5.14 *Winch.*

3.5.14.1 *Winch design.* The winch shall deliver a 45,000 pound rated line-pull at first layer on bare drum with 2 part line and 25,000 pound rated line pull at first layer on bare drum with single line. Winch shall conform to MS51354 modified for front or amidship mounting. Winch drive shall include torsional overload device.

3.5.14.2 *Wire rope and fittings.* Unless otherwise specified (see 6.2), each winch shall be furnished with wire rope not less than 280 feet in length. The wire rope shall be a continuous length without splice or junction, and shall be 3/4 inch diameter with a minimum breaking strength of 55,800 pounds and equipped with a clevis type fitting conforming to drawing 7071880.

3.5.14.3 *Winch controls.* Auxiliary vehicle throttle, clutch controls, power take-off, and winch controls shall be located adjacent to the winch and shall provide the winch operator with convenient and safe means of operating the winch.

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3.5.14.4 *Front winch assembly.* When specified (see 6.2), a front-mounted winch assembly of the lightest practicable weight shall be mounted on the M121A1 series trucks. The assembly shall include power takeoff, drive, control devices, wire rope, snatch-block, and mounting arrangements. The winch assembly shall be so constructed and arranged that the winch and its mounting can be removed as a unit. Vehicle furnished without a winch shall have a front bumper installed in that location. The torsional overload device of the front mounted winch shall prevent the exertion of a line pull in excess of 25,000 pounds maximum.

3.5.14.5 *Amidship mounted winch.* Winch(s), furnished for amidship mounting on M123A1C shall be equipped with level wind device(s), installed above the chassis frame, in back of cab, with subframe and mounting to withstand the strains imposed by operation of the winch. The torsional overload device of the amidship mounted winch shall prevent the exertion of a line pull in excess of 25,000 pounds maximum.

3.5.15 *Electrical system.* The electrical system furnished shall be a nominal 24 volt system, waterproof, with 4 12 volts batteries conforming to MS35000. When specified (see 6.2) the batteries furnished shall be in the dry charged or wet charged condition.

3.5.15.1 *Generating system voltage (alternator).* The alternator regulator voltage output shall be adjusted to 28 ± 0.2 volts (if not operating between 27.8 to 28.2 volts), when measured at the battery terminals. During vehicle operation, the battery charging indicator shall be observed to determine stability of voltage. At completion of the operating period, the voltage shall be re-adjusted, if required to 28 ± .2 volts.

3.5.15.2 *Lights and receptacles.* All the internal and external lights as specified on applicable drawings shall operate as required through all condition of vehicle operation.

3.5.16 *On vehicle equipment (OVE).* OVE shall be furnished in accordance with applicable equipment lists.

3.5.16.1 *Stowage provisions for OVE.* Suitable compartments tool box, brackets, or other stowage components shall be furnished to stow all OVE items.

3.5.16.1.1 *Stowage arrangements (Components).* Stowage arrangements (components) shall be located and installed on the vehicles to permit ready access to tools and equipment under loaded and operational conditions. The stowage or tool compartment shall be of waterproof construction, shall include means for draining and securing with a padlock conforming to MS35647-7.

3.5.16.2 *Brackets.* The following standard military brackets shall be included in the vehicle design as original equipment.

(a) Bracket, liquid container D6566675

(b) Bracket, pioneer tools 7346922

3.5.16.3 *Stowage space.* Stowage space shall be provided in vehicle design to permit subsequent (field) stowage of all OVM items listed on list of equipment troop installed.

3.5.17 *Radio interference suppression.* Unless otherwise specified (see 6.2) vehicle shall be radio interference suppressed for tactical vehicles in accordance with requirements class IIIA of MIL-STD-461.

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3.6 Painting, marking, and data plates.

3.6.1 *Painting.* Exterior and interior of vehicle, component assemblies, and parts which require painting, shall be prepared and painted in accordance with MIL-STD-193. Color shall be in accordance with Federal STD. No. 595, as specified for applicable service (see 6.2). Painted surfaces shall be free from sags, runs, and thin areas.

3.6.2 *Marking.* Vehicle marking shall be in accordance with MIL-STD-642, as specified for applicable service (see 6.2).

3.6.3 *Name, shipping, and service data plates.* The data plates shall meet the performance requirements of MIL-P-514.

3.7 Undercoating. Unless otherwise specified (see 6.2) vehicle shall be undercoated after priming is completed, using coating compound conforming to TT-C-520. Coating shall be applied to undersurfaces of cab, body and chassis metal to a thickness of 1/16 to 1/8 inch; except that drive shafts, drain holes, lubrication points, vents, engine crankcase, heavy castings, suspension components, and the underside of the engine compartment hood shall be kept free of coating material.

3.8 Servicing and adjustment. The contractor shall service the vehicle for operational use prior to delivery. Such servicing shall include, as a minimum the focusing of lights, the proper adjustment of the engine, electrical system, brake system, front wheel alinement and tire pressure. Unless otherwise specified, the chassis, engine and all running gear shall be serviced with lubricants of the proper grade for the climatic conditions at the delivery points. Engine cooling system shall be serviced with a solution of ethylene glycol and water in equal parts by volume.

3.9 Good driver plate. A good driver plate conforming to MS53050 shall be installed and so located in the driver's compartment as to be clearly visible and legible to the driver at all times.

3.10 Workmanship. The workmanship shall produce vehicles free from fabrication defects which would affect the appearance, functioning, or operating life of the vehicle or any of its components.

3.11 Air transportable. The vehicle shall be capable of air transportability in Phase III airborne operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Contractor's quality control system.* Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by the specification or other applicable documents referenced therein.

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4.1.2 *Government verification.* All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedure, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been taken or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 *Qualified products.* The contractor's inspection records shall be checked to determine contractor conformance to 3.3.1.

4.1.4 *Parts and components.* Parts, components and assemblies shall be inspected in accordance with the drawings listed on EPLS 8736670 M121A1, 8736671 M125A1, and 8736634 M123A1C, and Master Lists of Supplementary Quality Assurance Provisions 8736670 M121A1, 8736671 M125A1 and 8736634 M123A1C.

4.2 Preproduction and first production inspection.

4.2.1 *Preproduction inspection.* A preproduction vehicle (see 3.1) when furnished shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model to comply with any of the requirements specified shall be cause for rejection and refusal by the Government to conduct a retest until corrective measures satisfactory to the Government have been taken.

4.2.2 *First production vehicle inspection.*

4.2.2.1 *Special process inspection.* A special process inspection shall be conducted by the Government representatives during fabrication of the first production (see 3.2) vehicle of each model to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.2.1.1 *Review and evaluation.* Review and evaluation of records for processing procedure, inspection systems and plan of quality control, shall be conducted as a minimum on the following: material, welding, surface hardening, heat treating, and protective coating, etc.

4.2.2.1.2 *Chassis inspection.* The chassis shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.2.2 *Completed first production vehicle inspection.* The first production vehicle of each model when complete, shall be examined and given a 50 mile road test (with payload) by the contractor. The road test shall be in accordance with 4.4.2 and 4.4.3.6. The Government shall perform an inspection on the first production vehicle subsequent to contractors inspection to determine conformance to contract and specification requirements.

4.2.2.2.1 *Corrections.* Corrections found necessary as a result of above inspection shall be made by the contractor on the vehicle.

4.2.2.2.2 *Vehicle disposition.* Upon completion of inspection of the first production vehicle of each model it shall remain at the manufacturing facility as a production sample. The vehicle may be released for shipment after 120 days from start of production has elapsed or when the production run of the vehicle has

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been completed prior to the 120 days. If the quality level of the subsequent produced vehicle exceeds the acceptance standard, the first production vehicle may be released sooner at the discretion of the Government. The contractor shall service and maintain the vehicle during this period in accordance with the applicable documents for care and preservation while in storage.

4.2.3 *Production vehicle test.* After completion of the first vehicle inspection, a vehicle or vehicles shall be selected from the first months production and shall be subjected to examination (see 4.3.2) and tests as specified in table V at place of manufacturer. Subsequent to examination and tests, the selected vehicle shall be subjected to the 10,500 mile test specified in table II. These tests shall be performed by the Government, at a site approved by the Government, which will require a maximum of 90 days. Delays caused by vehicle break-down due to poor quality of workmanship or material, or failure of the contractor to comply with specifications or drawing requirements shall not be the basis for adjustment of the contract performance dates of delivery schedule or contract price. The contractor will expenditiously furnish repair parts, as required to adequately support the above tests. Delays caused by vehicle break-down and failure of the contractor to adequately support the vehicle with parts during the tests shall not be the responsibility of the Government.

TABLE II. *10,500 mile test*

Course	Mileage and speeds	Vehicle loading
Hard surface (highway)	3150 miles at variable speeds up to maximum	Highway
Level cross-country	4095 miles at speeds applicable to conditions of terrain	Cross-country
Hilly cross-country	3150 miles at speeds applicable to conditions of terrain	Cross-country
Belgian block	105 miles at speeds applicable to conditions of terrain	Cross-country

4.2.3.1 *Test failure.* Failure of the vehicle as the result of any deficiency of a workmanship or materials nature during or as a result of the 10,500 mile test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency. Any deficiency found during or as a result of 10,500 mile test shall be prima facie evidence that all vehicles already accepted prior to completion of the 10,500 mile test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Inspection provisions.

4.3.1 *Lot size.* An inspection lot shall consist of all vehicles of one type, from one day's production submitted at one time for acceptance inspection.

4.3.1.1 *Sampling for inspection.* For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100 percent inspect the first 20 vehicles to establish a process average (see 6.4) to allow normal sampling in accordance with MIL-STD-105.

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4.3.2 *Examinations.* Visual, dimensional, and primary functional examination shall consist of examination of the vehicle to assure conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in tables III and IV. the following constitutes a part of the classification of defects (see 6.5):

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.

(b) Any droplet that occurs at a static fit, metal to metal or gasket combination shall constitute a major defect.

TABLE III. *Major Defects AQL = 25 Defects/100 Units*

Characteristics	Def. No.	Major Defects	Method of Inspection
Steering mechanism	101.	Malfunction, leads (see 6.5-b)	Visual & functional
Engine	102.	Malfunction, leaks, (see 6.5-b), mounting improper	Visual & functional
Clutch	103.	Malfunction	Functional
Transmission	104.	Malfunction, leaks (see 6.5-b), mounting improper	Visual & functional
Prop, shafts	105.	Malfunction, misalinement	Visual & functional
Transfer	106.	Malfunction, misalinement (see 6.5-b) mounting improper	Visual & functional
Power take-off	107.	Malfunction, leaks (see 6.5-b)	Visual & functional
Cooling system components	108.	Malfunction, damage, leaks (see 6.5-b)	Visual & functional
Fuel system components	109.	Malfunction, damage, leaks (see 6.5-a)	Visual & functional
Engine governor	110.	Malfunction, seals, defective or missing	Visual & functional
Instrumentation, switches, warning, indicating and safety devices	111.	Malfunction, damage	Visual & functional
Ignition system components	112.	Malfunction, damage	Visual & functional
Electrical system components and circuits	113.	Malfunction, damage, lighting sequence improper	Visual & functional
Service and handbrakes	114.	Malfunction, leaks	Visual & functional
Axles	115.	Malfunction, leaks	Visual & functional
Exhaust system components	116.	Damage, exhaust leaks	Visual & functional
Suspension system components	117.	Damage, leaks	Visual

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TABLE III. *Major Defects AQL = 25 Defects/100 Units (continued)*

Characteristics	Def. No.	Major Defects	Method of Inspection
Operating units	118.	Overheating	Tactile (touch)
Speed	119.	Nonconformance	Functional
Windshield and windows	120.	Fogged, cracked, broken	Visual
Tires	121.	Damaged	Visual
Frame	122.	Structural and welding defects	Visual
Heater, defroster and blowers	123.	Malfunction, leaks	Visual & functional
Engine lubrication system	124.	Leaks, damage	Visual
Vehicle performance	125.	Erratic operation, malfunction	Visual & functional
Steering power assist mechanism	126.	Malfunction, leaks	Visual & functional
Front wheel limit valve and air supply valve	127.	Malfunction, component damage	Visual & functional
Suspension lock-out devices	128.	Malfunction	Functional
Wire rope size, length and type	129.	Improper size, length or type, defective	Visual
Springs and stops	130.	Spring sag under imposed load without proper clearance at stops	Visual

TABLE IV. *Minor Defects AQL = 150 Defects/100 Units*

Characteristics	Def. No.	Minor Defects	Method of Inspection
Controls	201.	Malfunction, improper clearance, adjustment or assembly	Visual & functional
Coolant	202.	Low or improper mixture	Visual & hydrometer
Lubricants	203.	Levels and types improper (all units)	Visual
Wheels and tire	204.	Improper size, type or mounting	Visual
Pulleys and fan	205.	Misalinement, improper mounting or clearance	Visual
Belts	206.	Tension improper, defective	Visual
Suspension system components	207.	Improper assembly or installation	Visual
Wiring or tubing	208.	Defective, improper assembly, installation, coding or protection	Visual
Brake system components	209.	Improper assembly, installation, clearance, protection, or control adjustment	Visual
Body, sheet metal, windshield seals, hardware, stowage boxes, and brackets	210.	Improper fits, assembly, or installation, defective weldments	Visual functional
Cushions, upholstery, canvas tops, curtains, seats and straps	211.	Damage, improper fits or installation, trim defects	Visual

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TABLE IV. *Minor Defects AQL = 150 Defects/100 Units (continued)*

Characteristics	Def. No.	Minor Defects	Method of Inspection
Misc. items and accessories	212.	Improper assembly or installation	Visual
Electrical system components	213.	Improper assembly or installation	Visual
Paint	214.	Application and color improper, defective	Visual
Steering system components	215.	Improper assembly, installation or clearance	Visual
Exhaust system components	216.	Improper assembly or installation	Visual
Cooling system components	217.	Improper assembly or installation	Visual
Fuel system components	218.	Improper assembly or installation	Visual
Lube fittings	219.	Defective, missing or improperly installed	Visual
Decals, markings, data and instruction plates	220.	Incomplete data, missing, improper locations and size, defective	Visual
Protective coatings	221.	Application, materials and coverage improper	Visual
Records, forms and publications	222.	Missing, improperly processed	Visual
Lubrication	223.	Improper	Visual
Vehicle cleanliness	224.	Improper	Visual
Pintle, towing shackles, access step, lifting and tie down devices	225.	Improper assembly or installation, malfunction	Visual & functional
Priming system	226.	Tapped holes improperly positioned for easy access	Visual
Hand control valve and stop light switch	227.	Improperly positioned for driver access	Visual

4.3.2.1 *Unclassified defects.* All defects having no effect on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's.

4.3.2.2 *Recurring major deficiency.* A major deficiency (see 6.6) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive lots. A major defect may be considered recurring when the Historical Inspection Records ("P" Chart or approved equivalent) reflect such a condition.

4.3.2.3 *Recurring minor deficiencies.* A minor deficiency (see 6.6) is recurring if it occurs more than twice in the same sample and/or when the defect occurs in four successive lots. Recurring minor deficiencies shall be cause for the entire lot and/or lots to be inspected for those recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance.

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4.4 Classification of tests.

(a) Acceptance test 4.4.1 through 4.4.1.22.

(b) Control tests 4.4.2 through 4.4.2.4.

(c) Comparison tests 4.4.3 through 4.4.3.9.

4.4.1 *Tests.*

4.4.1.1 *Acceptance tests.* To determine conformance to 3.5, each vehicle shall be operated for a distance of 50 miles or more by the contractor (without payload), and subjected to acceptance tests specified in Table V at place of manufacture. Vehicles shall be completely assembled and serviced with fuel, lubricants, and fluids as specified in lubrication chart. Prior to or during performance test vehicle(s) shall be examined as specified in 4.3.2. Vehicle performance shall be demonstrated on smooth, relatively level hard-surfaced roads. During these tests vehicle shall be operated in reverse gear for not less than 50 feet. After completion of tests the vehicle(s) shall be examined for evidence of fuel, lubricants, coolant, and fluid leaks (see 6.5).

TABLE V. *Location for tests*

Title	Paragraph References	Place of Mfg.	Government Proving Ground
Acceptance tests	4.4.1.1	X	
Seals check	4.4.1.2	X	X
Special kit checks	4.4.1.3	X	X
Balancing check	4.4.1.4	X	X
Brush guard check	4.4.1.5	X	X
Control checks	4.4.1.6	X	X
Seat check	4.4.1.7	X	X
Windshield, ventilation, and tops check	4.4.1.8	X	X
Fuel & air tanks check	4.4.1.9	X	X
Fuel & air lines check	4.4.1.10	X	X
Air cleaner check	4.4.1.11	X	X
Exhaust system check	4.4.1.12	X	X
Steering power-assist check	4.4.1.13	X	X
Compressed-air devices	4.4.1.14	X	X
Power package & power train test	4.4.1.15	X	X
Fuel system check	4.4.1.16	X	X
Transmission & components check	4.4.1.17	X	X
Front and rear axles check	4.4.1.18	X	X
Braking system check	4.4.1.19	X	X
Lower fifth wheel & equipment check	4.4.1.20	X	X
Winch, and controls check	4.4.1.21	X	X
Electrical system test	4.4.1.22	X	X
Control tests	4.4.2	X	X
Braking system operational test	4.4.2.2	X	X
Winch functional test	4.4.2.3	X	X
Stowed equipment check	4.4.2.4	X	X

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TABLE V. *Location for tests (continued)*

Title	Paragraph References	Place of Mfg.	Government Proving Ground
Comparison tests	4.4.3		X
Environmental tests	4.4.3.2		X
Cooling system test	4.4.3.3		X
Braking ability test	4.4.3.4		X
Cross-country operational tests	4.4.3.5		X
Highway operational tests	4.4.3.6		X
Water fording operational test	4.4.3.7		X
Gradeability test	4.4.3.8		X
Radio suppression test	4.4.3.9		X
Preservation, packaging and vehicle processing	4.5	X	

4.4.1.1.1 *Test failure.* If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall withhold acceptance until evidence has been provided by the contractor that corrective action has been taken.

4.4.1.2 *Seals check.* To determine conformance to 3.4.1, after testing has been completed the vehicle(s) shall be examined for leaking of bearing lubricants.

4.4.1.3 *Special kit and slave kit check.* To determine conformance to 3.4.2 through 3.4.2.3, and 3.4.16 when furnished the fording and arctic kits shall be installed on vehicle and checked for fit and completeness. The slave receptacle kit shall be checked for proper installation and damage.

4.4.1.4 *Balancing check.* To determine conformance to 3.4.3, during testing, each vehicle shall be checked, for wheel drumming or tramping.

4.4.1.5 *Brush guard check.* To determine conformance to 3.4.4, each vehicle shall be examined for proper installation of brush guards and completeness of guards.

4.4.1.6 *Control checks.* To determine conformance to 3.4.5, all controls, switches, and safety devices, shall be operated and checked for functional requirements.

4.4.1.7 *Seat check.* To determine conformance to 3.4.6, seats shall be checked for damage and function of adjustment control.

4.4.1.8 *Windshield, ventilation, and tops checks.* To determine conformance to 3.4.7, through 3.4.10.2, all assemblies and installations shall be checked for completeness as required for specified item.

4.4.1.9 *Fuel and air tanks checks.* To determine conformance to 3.4.11, tanks shall be examined for damage, installation, cleanliness, and leaks.

4.4.1.10 *Fuel, and air, lines check.* To determine conformance to 3.4.11.1, all lines shall be examined for damage, installation, and leaks.

4.4.1.11 *Air cleaner check.* To determine conformance to 3.4.12, air cleaner shall be checked for damage, and installation.

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4.4.1.12 *Exhaust system check.* To determine conformance to 3.4.13, the exhaust system shall be checked prior to and during operation for installation and leaks.

4.4.1.13 *Steering power-assist check.* To determine conformance to 3.4.14, prior to, during, and after the road test the steering power assist shall be checked for functional requirements.

4.4.1.14 *Compressed-air devices check.* To determine conformance to 3.4.15 through 3.4.15.4, the operable devices shall be operated and checked for functional requirements. Lines, reservoir, and air connectors shall be checked for proper installation, location, and damage.

4.4.1.15 *Power package and power train test.* To determine conformance to 3.5 and 3.5.2 through 3.5.3.3, the vehicle shall be operated, as specified and the power package and power train checked for functional requirements.

4.4.1.16 *Fuel system check.* To determine conformance to 3.5.4 and 3.5.4.1, the vehicle shall be placed on specified grades, and slopes, and fuel system checked for the supplying of fuel to engine.

4.4.1.17 *Transmission and components check.* To determine conformance to 3.5.5 through 3.5.5.2, during vehicle operation the transmission, power take-off mechanism, two speed transfer case, and full-torque power take-off mechanism shall be operated and checked for functional requirements.

4.4.1.18 *Front and rear axles check.* To determine conformance to 3.5.6 through 3.5.9, during vehicle operation the front and rear axles, and differentials shall be checked for functional requirements and cramping angle.

4.4.1.19 *Braking system check.* To determine conformance to 3.5.10 through 3.5.10.5, during operation of the vehicle all braking system components shall be operated and checked for functional requirements.

4.4.1.20 *Lower fifth wheel and equipment check.* To determine conformance to 3.5.13, the truck tractor shall be connected to a semi-trailer, and the fifth wheel checked for functional requirements. This check may be simulated, provided the procedure is approved by the commodity command.

4.4.1.21 *Winches, and controls check.* To determine conformance to 3.5.3.2 and 3.5.14, through 3.5.14.5, the winches and controls shall be operated and checked for functional requirements. For acceptance purposes reel and unreel line full length (less pull test) to assure operability of the unit.

4.4.1.22 *Electrical system test.* To determine conformance to 3.5.15, 3.5.15.1, and 3.5.15.2, the complete electrical system shall be checked for continuity from end to end of each circuit, all switches turned on and off for not less than 3 times and all light bulbs checked for lighting.

4.4.2 *Control tests.* The Government shall select at random one vehicle of each model per month or one out of each 100 vehicle for road test by the contractor, but not more than 2 vehicles in any given 30 day period. The vehicle shall be operated for a distance of not less than 75 miles with highway payload and all equipment installed, and subjected to applicable tests of 4.4.1.12 through 4.4.1.20 specified in table V.

4.4.2.1 *Failure.* If the vehicle selected fails to pass any of the control tests the Government inspector shall withold acceptance examination and testing on subsequent lots until such time as conditions causing the failure have been remedied.

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4.4.2.2 *Braking system operational test.* To determine conformance to 3.5.10.1, through 3.5.10.3, and 3.5.10.5, with a trailer attached and loaded to gross vehicle weight, the brakes and automatic breakaway device shall be operated to control stop and hold vehicle on specified grades and slopes.

4.4.2.3 *Winch operational test.* To determine conformance to 3.5.3.2 and 3.5.14 through 3.5.14.5 winches shall be subjected to pull test in accordance with applicable requirements for type specified. Size and length of wire rope shall be determined and shall be without splice of junction and other defects. Function of overload devices during tests shall be ascertained by line pull in excess of specified requirement. Controls and mechanism shall be observed for proper function, evidence of lubricant leaks or indication of other defects.

4.4.2.4 *Stowed material check.* To determine conformance to 3.5.16 through 3.5.16.3, all manufacturer and depot installed OVM shall be stowed on the selected vehicle. All items shall fit with proper clearance in the space provided. The equipment shall be removed after the test. The OVM, used shall be as specified in the contract; and the contractor shall be permitted to unpackage and repackage (for domestic shipment only) equipment used for test.

4.4.3 *Comparison tests.* The Government may select vehicles at any time during the contract period, and subject vehicle(s) to all tests specified in tables V and VI. Tests shall be conducted by the Government at a site selected or approved by the Government. Test mileage indicated shall be accumulated subsequent to 100 mile break-in operation at speeds up to maximum governed revolutions per minute.

TABLE VI. *5,000 mile comparison test*

Course	Mileage and speeds	Vehicle loading
Hard surface (highway)	1900 miles at variable speeds up to maximum	Highway
Level cross-country	1500 miles at speeds applicable to conditions of terrain	Cross-country
Hilly cross-country	1500 miles at speeds applicable to conditions of terrain	Cross-country
Belgian block	100 miles at speeds applicable to conditions of terrain	Cross-country

4.4.3.1 *Test failure.* Failure of any vehicle tested as specified in 4.4.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

4.4.3.2 *Environmental tests.* To determine conformance to 3.5.1, the vehicle, properly serviced and equipped, shall be subjected to the applicable tests in high and low temperatures.

4.4.3.3 *Cooling system test.* To determine conformance to 3.5.3.3, during these tests the vehicle shall be operated in such a manner, so as to assure cooling system is meeting operational requirements.

4.4.3.4 *Braking ability and breakaway device test.* To determine conformance to 3.5.10 through 3.5.10.5, the vehicle with rated payload and towed load shall be operated on relatively level hard surfaced road, in specified slopes, the braking system, and breakaway device operated and observed for performance requirements.

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4.4.3.5 *Cross-country operational tests.* To determine conformance to 3.5.11.1 and 3.5.11.2, the vehicle shall be operated as specified and vehicle observed for performance requirements.

4.4.3.6 *Highway operational test.* To determine conformance to 3.5.11.3 through 3.5.11.3.2, the vehicle with rated payload and towed load shall be operated at specified speeds, and observed for vehicle performance.

4.4.3.7 *Waterfording operational test.* To determine conformance to 3.5.11.4, the vehicle with rated cross-country payload and towed load shall be operated as specified and vehicle observed for performance requirements. Water contamination of wheel bearing lubricant not to exceed that specified in 3.4.1.

4.4.3.8 *Gradeability test.* To determine conformance to 3.5.12 through 3.5.12.2, the vehicle shall be operated as specified and vehicle observed for performance requirements.

4.4.3.9 *Radio suppression test.* To determine conformance to 3.5.17, the vehicle shall be subjected to a radio interference suppression test as specified.

4.5 Preservation, packaging and vehicle processing. Preservation and packaging of vehicle and equipment shall be inspected prior to shipment to assure conformance with the applicable requirements specified in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicles shall be processed for shipment and storage in accordance with MIL-V-62038 to the extent indicated on the Vehicle Preservation Data Sheets furnished by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general use by the United States Armed Forces in transporting personnel or cargo, or in providing for special tasks, during tactical military operations. When fitted with special equipment kits, the vehicles are intended for use under extreme or unusual conditions of climate, weather, terrain, and military service.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Preproduction vehicle if required (see 3.1).

(c) Special kits as required (see 3.4.2).

(d) Power take-off mechanism required (see 3.5.5.1).

(e) Automatic means if required (see 3.5.8).

(f) Wire rope and fittings (see 3.5.14.2).

(g) Front winch and level wind if required (see 3.5.14.4).

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(h) Batteries furnished dry or wet as required (see 3.5.15)

(i) Radio interference suppression if other than specified.

(j) Color for specific service as required (see 3.6.1).

(k) Marking for specific service as required (see 3.6.2).

(l) Underwriting not required (see 3.7).

(m) Level of packaging (see 5.1).

6.3 Safety precautions. Caution should be used in handling carbon dioxide fire-extinguisher cylinders. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control since physical injury is highly probable.

6.4 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process Average} = \frac{\text{Number of Defects}}{\text{Number of Vehicles Inspected}} \text{ X } 100$$

If the computed process average exceeds the specified AQL, 100% inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles in less than the specified AQL.

6.5 The following shall be used with the classification of defects for leaks:

a. Weep. Any evidence of fluid beyond the seal.

b. Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.

c. Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.

d. Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Definitions of recurring major and minor defects. The Government Inspection will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

Custodian:
Army—AT
Navy—MC
Air Force—85

Preparing activity
Army—AT
Project No. 2320-0547

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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